University of Florida, Department of Mathematics

Analysis PhD Exam, May 2018

Do six of eight. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

1. Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measurable spaces, $\mathcal{E} \subset \mathcal{N}$ and $f : X \to Y$. Show, if the σ-algebra generated by $\mathcal{E}$ contains $\mathcal{N}$ and $f^{-1}(E) \in \mathcal{M}$ for all $E \in \mathcal{E}$, then f is $\mathcal{M}$-$\mathcal{N}$ measurable.

2. Do two. In each case, give a brief justification for your answer.

(a) Give an example, if possible, of a measurable set $C \subset \mathbb{R}$ of positive Lebesgue measure that contains no (non-trivial) interval.

(b) Give an example, if possible, of a measure space $(X, \mathcal{M}, \mu)$ and a sequence of measurable functions $f_n : X \to \mathbb{C}$ that converge in measure, but not pointwise a.e.

(c) Give an example, if possible, of measure spaces $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ and a function $f : X \times Y \to [0, \infty)$ that is measurable with respect to the product measure σ-algebra $\mathcal{M} \otimes \mathcal{N}$, but for which

$$\int_X \int_Y f(x, y)\, d\nu\, d\mu \neq \int_Y \int_X f(x, y)\, d\mu\, d\nu.$$

3. Let $(X, \mathcal{M}, \mu)$ be a σ-finite measure space (μ a positive measure). Suppose $\mathcal{N}$ is a sub-σ-algebra of $\mathcal{M}$ and $\nu = \mu|_{\mathcal{N}}$ is σ-finite. Given $f \in L^1(\mu)$, show, by considering the mapping $\rho : \mathcal{N} \to [0, \infty)$ defined by $\rho(E) = \int_E f\, d\mu$, that there exists an $\mathcal{N}$-measurable g such that $g \in L^1(\nu)$ and

$$\int_E f\, d\mu = \int_E g\, d\nu$$

for all $E \in \mathcal{N}$. Determine g in the case $(X, \mathcal{M}, \mu)$ is the interval $[0, 1]$ with Lebesgue measure and $\mathcal{N} = \{\varnothing, [0, \frac{1}{2}), [\frac{1}{2}, 1], [0, 1]\}$.

4. Show $f : [0, \infty) \to \mathbb{R}$ defined by $f(x) = (1 + x)^{-2}$ is integrable (with respect to Lebesgue measure on $[0, \infty)$). Determine

$$\lim_{n \to \infty} \int_0^\infty \frac{\sin(x/n)}{(1 + (x/n))^n}\, dx.$$

5. Prove, if X is a Banach space and M is a finite dimensional subspace of X, then there is a closed subspace N of X such that $M \cap N = (0)$ and for each $x \in X$ there exist $m \in M$ and $n \in N$ such that $x = m + n$.

6. Let $(X, \mathcal{M}, \mu)$ be a measure space and suppose $1 \leq q < p < \infty$. Show, if $L^p(\mu) \subset L^q(\mu)$, then there is a constant κ such that if $E \in \mathcal{M}$ and $\mu(E) < \infty$, then $\mu(E) \leq \kappa$.

7. Suppose $E \subset \mathbb{R}$ is (Lebesgue) measurable with positive measure. Show

$$E - E := \{x - y : x, y \in E\}$$

contains an open interval about 0.

8. Recall c_{00} is the vector subspace of $\ell^\infty(\mathbb{N})$ consisting of sequences $a = (a_n)$ that are eventually zero (meaning there is an N, depending on a, such that $a_n = 0$ for $n > N$). Is there a norm on c_{00} that makes c_{00} a Banach space?